

DCVLR: Data Curation for Vision Language Reasoning Competition

Oussama Elachqar^{*,2} Benjamin Feuer^{*,1,2} Neha Hulkund^{*,4} Rohun Tripathi^{*,3}
 Rohun Agrawal^{†,8} Thao Nguyen^{†,5} Nimrod Shabtay^{†,15,16} Vishaal Udandarao^{†,6,7}
 Xiaohan Wang^{†,1} Stefan Webb^{†,2} Yuhui Zhang^{†,1} Sara Beery^{‡,4} Georgia Gkioxari^{‡,8}
 Paul Liang^{‡,4} Ludwig Schmidt^{‡,1} Saining Xie^{‡,9} Serena Yeung-Levy^{‡,1} Michael
 Buriek^{§,12,13} Pavel Bushuyeu^{§,12} Dimitrios Damianos^{§,10} Eleni Kasoura^{§,10,11} Vas-
 silis Katsouros^{§,10} Vikas Kumar^{§,12} Igor Molybog^{§,12} Georgios Paraskevopoulos^{§,10}
 Yosub Shin^{§,12} Boris Sobolev^{§,12,14} Samuel Watson^{§,12} Haoyang Xu^{§,12} Emmanouil
 Koukoumidis (Manos)^{α,2}

^{*}Lead Organizer ^αSponsor [†]Helping Organizer [‡]Advisor [§]Participant

¹Stanford University ²Oumi.AI ³AllenAI ⁴MIT ⁵Meta ⁶University of Tübingen ⁷University
 of Cambridge ⁸Caltech ⁹NYU ¹⁰Institute for Language and Speech Processing, Athena Research
 Center ¹¹Department of Informatics, University of Athens ¹²University of Hawaii at Manoa ¹³PwC
¹⁴Cisco ¹⁵IBM Research ¹⁶Tel Aviv University

DCVLR_NEURIPS@GOOGLEGROUPS.COM

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Abstract

At NeurIPS 2025, we presented the first data-centric, open-data and open-models competition for data curation in vision-language reasoning. The competition focused on improving visual reasoning capabilities of vision-language models (VLMs) through data curation under a fixed supervised-finetuning (SFT) configuration and budget. Submissions were evaluated by fine-tuning a fixed VLM [Clark et al. \(2026\)](#); [Bai et al. \(2025\)](#) on the curated datasets and measuring performance on reasoning benchmarks of multiple-choice visual reasoning questions spanning diverse sources [Zhang et al. \(2025\)](#); [Shabtay et al. \(2024\)](#). The competition provided all necessary resources, including fine-tuning scripts, evaluation code, and baselines. Additionally, as a first of competitions of this kind, organizers also ran the entire SFT and evaluation pipeline on behalf of the contestants, thereby removing a key barrier to entry present in previous data-centric competitions. The result was the largest data-centric competition to date, with 130 signups and 20 submissions. DCVLR centered data curation as the key variable for scientific progress in VLM adaptation, highlighting its importance for real-world applications such as education, biomedicine, and scientific reasoning.

Keywords: data curation, vision-language models, visual reasoning, instruction tuning, data-centric AI, multimodal learning, competition, benchmark

1. Introduction

Visual reasoning has emerged as a central focus in developing next-generation AI [Bai et al. \(2025\)](#); [Singh et al. \(2025\)](#); [Marsili et al. \(2025\)](#); [Hudson and Manning \(2019\)](#); [Yue et al. \(2024\)](#); [Lu et al. \(2024\)](#). A visual reasoning task is any task that requires interpreting and

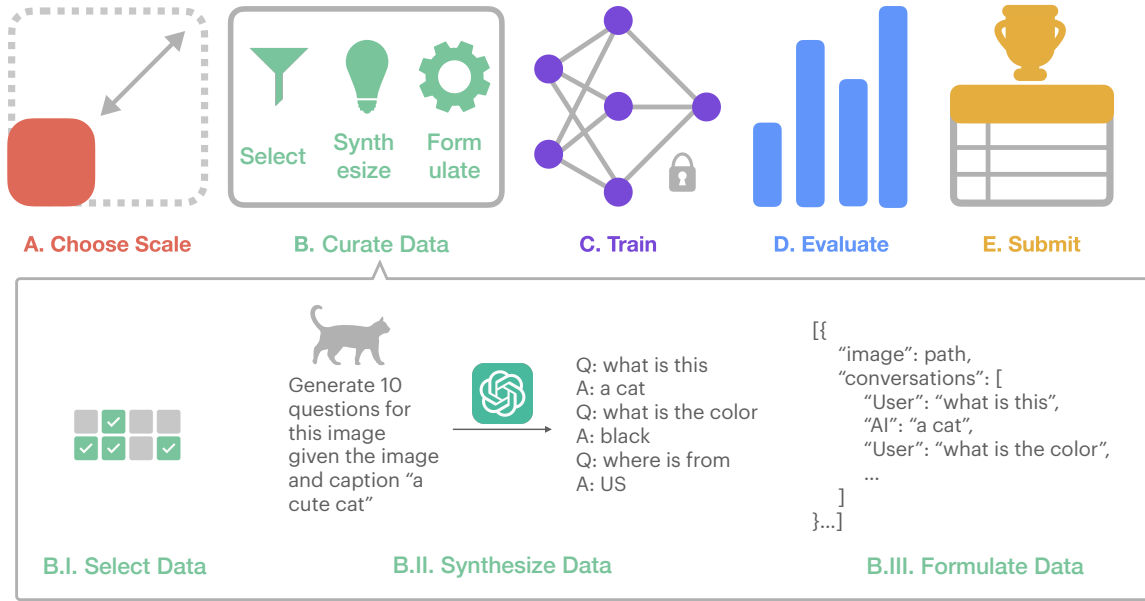


Figure 1: Overview of the DCVLR competition. Participants submitted curated instruction-tuning datasets (up to 10K samples) of their own construction; organizers fine-tuned a fixed, undisclosed vision-language model on each submission and scored it on a suite of held-out visual reasoning benchmarks.

reasoning over visual inputs, and that tends not to saturate with increased model size or pre-training data alone, but instead benefits from improved multi-modal grounding and higher-quality compositional reasoning over image-text interactions. Visual reasoning requires systems to infer causal relationships, interpret spatial configurations, and extract implicit information from visual stimuli. This cognitive leap—from seeing to reasoning—represents a crucial frontier for AI, exemplified in autonomous vehicles that must anticipate a child following a ball, or diagnostic systems that must integrate subtle radiological patterns to form coherent medical hypotheses.

Vision-Language Models (VLMs) like GPT-5 [Singh et al. \(2025\)](#) and Claude-4.7 [Anthropic \(2026\)](#) have shown promising reasoning abilities—for example, interpreting visual scenes or answering complex factoid questions given an image. Yet the question remains: how can we push these capabilities even further?

The Data Curation for Vision Language Reasoning (DCVLR) challenge explored this question through a rigorously controlled data-centric paradigm [Gadre et al. \(2023\)](#); [Li et al. \(2024\)](#). In this challenge, participants were tasked with curating datasets that improved an undisclosed VLM’s visual reasoning capabilities. Participants were encouraged to curate data via any strategy that was most effective. The competition imposed only two restrictions: 1) the submitted set had to contain fewer than 10K samples, and 2) no manual annotation was allowed. By fixing the model architecture and training protocol, we isolated and examined the causal relationship between dataset characteristics and reasoning emer-

gence. This approach inverts the traditional model-centric research paradigm, recasting data curation as the primary scientific variable rather than a mere preprocessing step.

Researchers aiming to explore this problem often face significant computational constraints, particularly limited access to adequate compute resources. We removed the computational barriers that typically constrain data curation innovation by providing GPUs to train and evaluate the models. This democratized participation and we attracted a diverse cohort of 130 teams and 20 final submissions spanning students and academic researchers to industry professionals, consistent with participation patterns in analogous data-centric competitions and the rapidly expanding VLM research community. Further to motivate performance and novelty, we established a prize pool of \$4,250 USD across three categories.

Fundamentally, this competition reframed data curation as a scientific discipline with its own theoretical questions, methodological approaches, and evaluative criteria. Our competition asks: what data properties catalyze reasoning capabilities in neural systems? By analyzing successful submissions, we extracted principles for effective data curation that transcend specific architectures. These principles offer not merely heuristics but a broader framework for understanding how to systematically instill reasoning abilities in multi-modal systems across domains.

Our best performing curation strategy focused on quality and difficulty of the task rather than quantity, demonstrating previously well-studied quality versus quantity trade-offs [Ye et al. \(2025\)](#); [Abdin et al. \(2024\)](#). The competitors combined two sets to create their high quality 1K training set: a selected set of 900 samples chosen by optimizing for example difficulty and a small set of 100 samples with annotated reasoning traces from an expert model [OpenAI \(2023\)](#). The combined set of 1K samples outperformed other submissions (which used up to 10K samples), demonstrating how a small set of hard high quality samples outperform larger data pools on this task.

2. Related Work

The most comparable existing competitions are DataComp [Gadre et al. \(2023\)](#) and DataComp-LM [Li et al. \(2024\)](#). DataComp focuses on curating data to train better contrastive vision-language models (e.g., CLIP [Radford et al. \(2021\)](#)), while DataComp-LM focuses on language model (LM) training. In both, participants are provided with a massive dataset and are asked to filter and select data to optimize downstream performance.

DCVLR was an entirely new competition centered on data curation for visual reasoning. It offered a data-centric framework for improving the reasoning capabilities of vision-language models (VLMs) by tasking participants with generating high-quality training data, and differed from these prior efforts in three key ways:

- 1. Focus on data creation, not just selection.** Unlike DataComp and DataComp-LM, which restrict participants to selecting from preexisting datasets, we allowed users to bring their own data and only provided references to related datasets that could be used to seed it. Participants were allowed to use any method—retrieval, filtering, augmentation, or synthesis—to generate their data. This design offered far greater creative freedom and encouraged exploration of novel data construction strategies.

- 2. Targeting reasoning, not general capability.** While prior competitions often emphasize general model performance, our competition was specifically designed to probe

visual reasoning. By focusing on this narrower yet critical capability, we aimed to derive actionable insights into how to steer foundation models toward more sophisticated reasoning. These findings can inform downstream applications such as biomedical image understanding, visual tutoring systems, or causal scene interpretation.

3. Fine-tuning over pre-training. Rather than evaluating large-scale pretraining efforts, we focused on fine-tuning existing, high-performing VLMs using curated instruction-tuning data. This significantly reduced computational requirements, making the competition more accessible to academic teams and smaller research groups. It also aligned better with real-world scenarios, where practitioners adapt foundation models to specialized tasks using modest resources.

3. The DCVLR Competition

Rather than focusing on novel architectures or training techniques, DCVLR put reproducible methods for data curation front and center. Participants were tasked with curating up to 10,000 instruction-tuning examples using any strategy and submitting them. The organizers handled all training and evaluation, so participants never needed to run large-scale training themselves.

3.1. Evaluation

The DCVLR evaluation suite consisted of ten benchmarks spanning diverse forms of multimodal reasoning, 10 benchmarks spanning mathematical reasoning, scientific comprehension, and spatial understanding. At the time of the competition, four of the benchmarks were known to participants:

- VMCBench-DEV (Zhang et al., 2025): visual reasoning benchmark that unified 20 existing VQA datasets into a consistent multiple-choice format by simulating errors from different perspectives
- OlympiadBench. (He et al., 2024): bilingual (English and Chinese) competitive Olympiad visual physics and mathematical reasoning questions, each with annotated step-by-step solutions.
- LiveXivVQA (Shabtay et al., 2024): visual question-answer pairs from figures, charts, and tables in recent scientific arXiv papers
- Omni3DBench. (Marsili et al., 2025): non-templated spatial queries on diverse real-world scenes, requiring 3D object grounding and reasoning about distances and dimensions, with four question types: yes/no, multiple choice, counting, and float-point calculations evaluated via mean relative accuracy.

The other six (physical visual understanding) tasks from the Yale Physics Benchmark (Feng et al., 2025) were held-out until final evaluation.

Participants were asked to fine-tune vision-language models from the following options: Qwen2.5-VL-7B (Yang et al., 2024), Molmo (Deitke et al., 2024), or InternVL (Chen et al., 2023). At the time of the competition, the exact evaluation model was deliberately not given

to participants to avoid exploitation of model-specific quirks or formatting practices in data curation. The goal was to ideally find a generalizable visual-reasoning data pool across model families. Final performance was given as the average accuracy over all benchmarks.

3.2. Baselines

We provided 5 key baselines as points of reference for the competition participants:

VL: Visual Learning through Synthetic Generation. This baseline was a multi-modal adaptation of the “Simple Scalable Test-Time Compute” paper (Muennighoff et al., 2025), which demonstrated that a small, carefully selected set of reasoning traces could yield strong performance gains. The strategy produced a curated collection of 1,000 high-quality vision-language pairs, generating synthetic data points that targeted common reasoning gaps in existing datasets. We used Claude Sonnet 3.7 to generate Python code that rendered contextually appropriate images (diagrams, geometric figures, charts), which were in turn used to convert text-only reasoning problems into multimodal examples.

LIMO: Language-Informed Multimodal Optimization. This baseline was a VL adaptation of the “Less is More for Reasoning” (LIMO) paper (Ye et al., 2025), which found that models pretrained on broad mathematical knowledge need only a small number of carefully constructed instruction-tuning examples to unlock strong reasoning. The strategy took a language-informed approach to multimodal data synthesis, leveraging Claude Sonnet 3.7 to create contextually rich and semantically coherent vision-language pairs, similarly to the VL baseline. This was also a 1,000-sample synthesis. Both VL and LIMO showed weaker transfer to LiveXivTQA compared to the filtration baselines, suggesting that synthesis alone at small scale does not easily target that distribution.

Multimodal Open R1: Reasoning-focused Filtration. This baseline was derived from the open-source reimplementation of DeepSeek-R1’s data generation methodology (DeepSeek-AI et al., 2025), extended to the multimodal setting. It applied reasoning-focused filters to identify data points that specifically enhance multimodal reasoning capabilities, building upon the Open R1 methodology, which emphasizes the importance of reasoning chains and step-by-step problem solving in multimodal contexts. The filtering process prioritized samples that demonstrated clear reasoning patterns and complex visual-textual interactions. The source dataset (`multimodal-open-r1-8192-filtered-mid-ic` on Hugging Face) contains chain-of-thought supervision in R1’s characteristic long-form reasoning style, making it potentially strong for tasks requiring explicit step-by-step derivation—though this style also did not generalize as well to the text-heavy LiveXivTQA benchmark.

MM MathInstruct: Mathematical Reasoning at Scale. The MM MathInstruct strategy focused on mathematical reasoning capabilities, filtering from a large corpus of mathematical instruction data (Wang et al., 2025) to identify the most effective samples for enhancing quantitative reasoning in vision-language models. This approach is particularly valuable for applications requiring numerical analysis, geometric reasoning, and mathematical problem-solving across visual and textual domains. It targeted the same OlympiadBench and physics-style reasoning that our competition benchmarks probed, but the results revealed a consistent ceiling: OlympiadBench remained uniformly challenging, with both the baseline Qwen-7B model and fine-tuned models achieving only around 13%

accuracy—suggesting that it requires reasoning capabilities beyond what current data curation strategies can address.

Walton Multimodal Cold Start: Bootstrap Learning Strategy. This strategy addressed the challenging problem of bootstrap learning in multimodal contexts. It was derived from a larger multimodal cold-start corpus (the “Walton” dataset (Oumi AI, 2025)) and filtered by the organizers into a clean 10K set. The Walton strategy achieved a 37.6% relative improvement (from 58.0% to 79.8%) on LiveXivTQA, bringing the fine-tuned model close to Claude Sonnet’s performance on that benchmark. The winning team extended this strategy to produce their final result.

For the sake of completeness, we include the model scores from those baseline runs in this proceedings; however, because we, as organizers, inevitably had access to more information about the competition, we ranked our contestants and reported results to them in relation to the **non-fine-tuned base model**, which was Qwen2.5-VL-7B-Instruct.

3.3. Participation

DCVLR ran as an open prize competition from June through October 2025, organized by Oumi.AI and Lambda.AI in collaboration with researchers from Stanford, MIT, NYU, UW, Caltech, AI2, the University of Tübingen, and Tel Aviv University, among others. Registration and evaluation were coordinated through the competition website¹, with technical support provided via a public Discord server and a GitHub starter kit distributed through the Oumi repository.

Engagement spanned a broad funnel from passive interest to full submission. We recorded 130 signups to the competition mailing list and 78 team registrations, of which 20 produced a final dataset submission that was evaluated on our held-out benchmarks. To our knowledge, this makes DCVLR one of the largest data-centric competitions in the vision-language setting to date. Participating teams represented a mixture of academic research groups, industry labs, and independent researchers, consistent with the democratized-access design of the competition: because the organizers handled all fine-tuning and evaluation on shared infrastructure donated by Lambda Labs, contestants only needed modest local resources to prototype and iterate on curation strategies.

A total prize pool of \$4,250 USD was awarded across three categories: \$3,000 for first place, \$1,000 for the Innovation in Data Curation (IDC) prize, and \$250 for Honorable Mention. Prize-winning teams were additionally invited to present their methods at the NeurIPS 2025 workshop and to co-author this proceedings. First place was awarded to the University of Hawaii at Manoa (team 3).

Because submissions were open-format BYOD (bring-your-own-data), every submission required automated cleaning before evaluation in order to keep training compute from being wasted on malformed or contaminated data. We processed all submissions through the purpose-built **data-preproc**² utility, which normalized input schemas and image sizes, applied text- and image-toxicity filters, and deduplicated submissions against both internal training corpora and our evaluation benchmarks (including the held-out Yale Physics splits). Toxicity and format normalization removed on the order of 1% of samples per submission on

1. <https://dcvlr-neurips.github.io/>

2. <https://github.com/penfever/data-preproc/>

average, while benchmark deduplication removed between roughly 5% and 25% of submission content depending on the team’s data sources—an empirical confirmation that overlap with public benchmarks is a non-trivial failure mode in data-centric vision-language work, and one that a well-run competition must explicitly control for.

4. Competition Results

4.1. First Place Solution

The University of Hawaii at Manoa, the 1st place finisher of DCVLR, team submitted a compact 1,000-example instruction-tuning dataset constructed by applying difficulty-based filtering to a strongly aligned base corpus. They started from the organizer-provided Walton Multimodal Cold Start dataset, chosen as the base because it exhibits strong distributional overlap with LiveXivTQA — the heavily weighted, text-rich benchmark in the DCVLR evaluation suite — and it is well aligned with the pretrained Qwen2.5-VL-7B-Instruct base model used for fine-tuning. As such, a small data budget could reinforce regions of the task distribution that the model already represents effectively. The primary curation signal was example difficulty, defined as the inconsistency of the base model’s answers under stochastic decoding: for each Walton example we ran Qwen2.5-VL-7B-Instruct sixteen times at temperature 0.7 and top-p 0.9, kept examples on which the model was correct on fewer than half of the rollouts, and combined the resulting subset with a small fraction ($\approx 10\%$) of CoSyn-400k examples whose reasoning traces had been rewritten with GPT-4o into longer, more explicit chain-of-thought form. Lightweight token-length filtering and deduplication produced the final 1,000-sample submission.

4.2. Second Place Solution

Team AFIE (Ant Group and HKUST), the 2nd-place finisher in the 1st edition of DCVLR, submitted a four-stage data curation framework titled Multi-Perspective Evaluation and Refinement, aimed at producing high-quality, balanced, and reasoning-intensive multi-modal training data. The team aggregated samples from eight heterogeneous vision-language sources — including LLaVA-OneVision, Multimodal-Open-R1, VLAA-Thinking-SFT, Vision-R1-Cold, MM-Eureka-Distill, R1-Onevision, OpenVLThinker-sft-iter3, OlympiadBench, and BMMR — spanning four task categories (chart QA, document QA, general knowledge QA, and mathematical reasoning QA). In the first stage, they applied stratified balancing jointly over task category and estimated difficulty (easy / medium / hard), intentionally up-sampling hard examples while preserving proportional representation of easier tiers, so the resulting mixture would drive deep reasoning without collapsing into a narrow, over-represented regime.

The second stage introduced a multi-dimensional reject-sampling layer in which four independent LLM/MLLM judges scored every candidate along factual correctness, logical soundness, completeness, and conciseness, with a sample required to clear predefined thresholds on all four axes to pass. Rather than discarding failures, the team routed them into a rejection pool for stage three, where per-sample diagnostic scores and textual feedback from the judges drove a targeted repair loop — automated answer revision to fix factual or logical flaws, and reasoning reconstruction to expand, reorganize, or simplify multi-step

chains — after which refined items were re-submitted to the same four-judge gate and, only if they passed, merged back into the curated pool. The final stage used the combined accepted-plus-refined dataset for SFT and benchmark evaluation.

4.3. Third Place Solution

AthenaRC, the third place finisher of DCVLR, submitted a synthetic multimodal instruction synthesis pipeline, where textual instructions are synthesised based on input images. Starting from the Cauldron collection, they sampled images from the 50 available subsets spanning VQA, OCR, charts, diagrams, spatial reasoning, and mathematical reasoning tasks. They then adapted a Magpie-style self-synthesis framework to the vision-language setting using Qwen2.5-VL-7B-Instruct model. Conditioned on the input image and system prompts, the model first generated user instructions grounded in visual content and subsequently produced paired assistant responses. To improve dataset quality, they applied multiple post-processing stages including sanitization to remove LLM artefacts, formatting validation, duplicate removal, and reward-based filtering. Each generated instruction-response pair was scored using the OpenAssistant DeBERTa-v3 reward model, and only positively scored samples were retained, with the final dataset consisting of the top-ranked 10,000 examples. They found that prompt steering and careful cleaning were especially important, as vision-language models tend to default toward generic captioning behavior without explicit guidance.

5. Results

5.1. Baseline Leaderboard

As shown in Table 1, during our baseline dataset evaluations, performance varied considerably within a fixed model scale, with effective data curation offering a decisive advantage on certain benchmarks, particularly those related to chart and infographic interpretation. This distinction turned out to also be important on our final leaderboard.

5.2. Final Leaderboard

During our baseline dataset evaluations, model capability reliably increased as a function of dataset scale; s1.1-VL and LIMO-vis underperformed compared to larger datasets such as Walton-R1. However, the submission from the first place team clearly demonstrated that scale alone is not a prerequisite for visual reasoning; careful sub-selection from a larger high-quality data pool can produce very strong results in this domain. The large spread of team performance is also noteworthy, considering the relatively small datasets used for the competition, suggesting again that effective data curation can be highly impactful in this regime.

5.3. Why is Walton-R1 a Good Data Source?

Given our findings, it is natural to wonder why the Walton-R1 data performs so well. While we cannot answer this question definitively, we can offer some plausible explanations; compared to the other baseline data sources, which mostly had open-ended responses, Walton-

Baseline	VMC	O3D	OlyB	LXV	YPhys	Avg
MM-Open-R1	77.1	33.3	09.7	66.5	07.1	38.7
s1.1-VL	78.8	33.9	10.2	64.6	05.6	38.6
Molmo-7B-O [†]	64.6	25.1	01.0	35.5	02.6	25.8
MathInstruct-R1	80.0	30.8	07.1	65.8	08.5	38.4
LIMO-vis	79.1	33.5	10.8	65.0	07.9	39.3
Qwen2.5-VL-7B (Base)	79.1	35.1	13.2	66.6	09.4	40.7
Walton-R1	78.7	34.6	13.1	79.2	10.5	43.2

Table 1: DCVLR baseline comparison across reasoning-oriented fine-tunes of Qwen2.5-VL-7B-Instruct, plus Molmo-7B-O for reference. **Abbreviations:** VMC = VMCBench (dev); O3D = Omni3DBench; OlyB = OlympiadBench; LXV = LiveXiv (unweighted average of the TQA and VQA splits); YPhys = Yale Physics (average of the atomic, electro, mechanics, optics, quantum, and statistics subsets); Avg = macro-average across the five reported benchmarks. [†]Molmo-7B-O was not evaluated on LiveXiv-TQA, so its LXV entry reflects VQA only. **Findings:** The Walton-R1 SFT mix remained the only curation to improve over the Qwen2.5-VL-7B-Instruct base on the macro-average (43.2% vs. 40.7%, +2.5 absolute); this gap is considerably narrower than a previously reported +10.2 because the earlier spreadsheet reported LiveXiv-TQA-only for most rows but TQA+VQA averages for Walton and MM-Open-R1, artificially inflating those two scores. Under a consistent TQA+VQA average, Walton’s advantage comes entirely from LiveXiv (79.2% vs. 66.6%, +12.6), while VMC, Omni3DBench, OlympiadBench, and Yale Physics are essentially tied with the base model. Every other reasoning-SFT variant (LIMO-vis, s1.1-VL, MathInstruct-R1, multimodal-open-R1) under-performed the base on the consistent average, suggesting that naive CoT distillation into a vision-language model can degrade broad VQA capability more than it helps symbolic reasoning. OlympiadBench and Yale Physics remained the hardest benchmarks for every model (all $\leq 14\%$ and $\leq 11\%$ respectively), confirming substantial headroom for future data-curation work on olympiad-level symbolic and physical reasoning.

R1 had a large amount of multiple-choice questions; this may have helped the model on the LiveXiv benchmark, which is multiple choice. Walton-R1 is also the only dataset which contains a significant amount of Chinese (around 11%); since OlympiadBench is dual-lingual, the Chinese content may have helped prevent the regression on that benchmark we observed with other data sources.

6. Conclusion

In this work, we reframed data curation as a scientific discipline with its own theoretical questions, methodological approaches, and evaluative criteria. By fixing the model and training pipeline, the DCVLR challenge isolated the effect of data, enabling a clearer view of how visual reasoning emerges from curated supervision. A key takeaway aligns with findings

Rank	Team	Accuracy (%)	Δ vs Baseline (%)
1	University of Hawaii at Manoa – team 3	46.12	+7.28
2	AFIE	42.13	+3.29
3	AthenaRC	41.81	+2.98
4	ZhuYun – PJLab	38.84	+0.01
5	CoreData	37.39	−1.44
6	Blackwell	37.11	−1.72
7	MICV	34.83	−4.00
8	KDDI Research	20.52	−18.31
–	<i>Baseline</i>	<i>38.83</i>	-

Table 2: DCVLR Final Leaderboard: NeurIPS 2025 Competition Results. Models were trained and evaluated by the organizers; accuracy is reported on a held-out benchmark and Δ denotes improvement over the provided baseline, Qwen2.5-VL-7B. Qwen2.5-VL-7B scores differ slightly between the original baseline evaluations and the final run because of differences in evaluation dataset composition between the baseline and final leaderboards.

from previous works (Ye et al., 2025; Abdin et al., 2024; Zhou et al., 2023): even small amounts of high-quality, targeted data can significantly improve reasoning performance. Our results suggests that progress in the visual reasoning domain may depend more on identifying the right data than simply scaling quantity.

Looking forward, our proposed paradigm can be extended to new settings. Applying data-centric curation to robotics could shed light on how multimodal experience shapes decision-making, while multilingual extensions would test how reasoning transfers across languages and cultures.

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